

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
H H T H H H T H H T
Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
2. Out of 50 students, 42 passed an exam.
Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
Find the MLE for the probability that a bulb is defective.
4. A biased die is rolled 60 times. The number 6 appears 18 times.
Estimate the probability of rolling a 6 using MLE.
5. Given:
Input (x) = 4
Weight (w) = 2
Actual Output = 10
Prediction:
 $y = w \times x$
 1. Find the predicted output.
 2. Calculate the error.
 3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Code of Conduct:

I Arya. B. Nair - 192424133 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

ASSESSMENT TOOL 1

Answers

1.

Given:

H H T H H T H H T H

Number of Heads = 7

Total tosses = **10**

For a Bernoulli distribution, the MLE is:

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total Tosses}}$$
$$\hat{p} = \frac{7}{10} = 0.7$$

2. Student Pass Probability

Given:

- Total students = 50
- Passed = 42

MLE:

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total Tosses}}$$
$$\hat{p} = \frac{42}{50} = 0.84$$

3. Defective Bulbs

Given:

- Total bulbs = 100
- Defective = 8

MLE:

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total Tosses}}$$
$$\hat{p} = \frac{8}{100} = 0.08$$

4. Biased Die

Given:

- Total rolls = 60
- Number of sixes = 18

MLE:

$$\hat{p} = \frac{\text{Number of Heads}}{\text{Total Tosses}}$$
$$\hat{p} = \frac{18}{60} = 0.30$$

5. Gradient Descent Problem

Given:

- Input $x = 4$
- Weight $w = 2$

- Actual Output = 10
- Prediction formula:

$$y = wx$$

(1) Predicted Output

$$y = 2 \times 4 = 8$$

(2) Error

Using the simple prediction error:

$$\begin{aligned}\text{Error} &= \text{Actual} - \text{Predicted} \\ &= 10 - 8 = 2\end{aligned}$$

(3) Updated Weight

Gradient descent update rule:

$$w_{\text{new}} = w - \eta \times \text{gradient}$$

where

- $w = 2$
- Learning rate $\eta = 0.01$
- Gradient = 4

$$\begin{aligned}w_{\text{new}} &= 2 - (0.01 \times 4) \\ &= 2 - 0.04 \\ &= 1.96\end{aligned}$$

ASSESSMENT -1

COL

Name: Anya . B. Nair

Reg No: 192424133

Course: MLAO4 - Deep Learning

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are :

H H T H H H T H H T

Using maximum Likelihood Estimation, estimate the probability of getting Heads (p).

Given outcomes :

10 Tosses —

H H T H H T H H T H

No. of heads = 7

Total tosses = 10

for Bernoulli's Distribution

$$\hat{p} = \frac{\text{Number of heads}}{\text{Total Tosses}}$$

$$= \frac{7}{10} = 0.7$$

$$\boxed{\hat{p} = 0.7}$$

2. Out of 50 students, 42 passed an exam. Assuming a Bernoulli distribution, find the MLE of the probability that a student passes

Given:

$$\text{Total students} = 50$$

$$\text{Passed} = 42$$

MLE

$$\hat{p} = \frac{\text{Passed}}{\text{Total students}} = \frac{42}{50} = 0.84$$

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective. Find the MLE for the probability that a bulb is defective.

Given:

$$\text{Total bulbs} = 100$$

$$\text{Defective} = 8$$

MLE,

$$\hat{p} = \frac{8}{100} = 0.08$$

4. A biased die is rolled 60 times. The number 6 appears 18 times. Estimate the probability of rolling a 6 using MLE.

Given :

$$\text{Total rolls} = 60$$

$$\text{Number of } \left. \begin{array}{l} \text{sixes} \end{array} \right\} = 18$$

MLE,

$$\hat{p} = \frac{18}{60} = 0.30$$

5. Given :

$$\text{input } (x) = 4$$

$$\text{weight } (x) = 2$$

$$\text{Actual output} = 10$$

Prediction:

$$y = w \times x$$

1. Find the predicted output
2. Calculate the error
3. If the gradient is 4 and learning rate 0.01, find the updated weight.

Given :

$$\text{Input } x = 4$$

$$\text{Weight } w = 2$$

$$\text{Actual output} = 10$$

Formula :

$$y = wx$$

(1) Predicted Output

$$y = 2 \times 4 = \underline{8}$$

(2) Error

$$\text{Err} = \text{Actual} - \text{Predicted}$$

$$= 10 - 8 = \underline{2}$$

(3) Updated Weight

$$w_{\text{new}} = w - \eta \times \text{gradient}$$

$$w = 2$$

$$\text{Learning rate } \eta = 0.01$$

$$\text{Gradient} = 4$$

$$w_{\text{new}} = 2 - (0.01 \times 4)$$

$$= 2 - 0.04$$

$$= 1.96$$

$$\text{Updated weight} = 1.96$$